

PROJECT REPORT

[Ration Distribution Management System]

Project ID: P16

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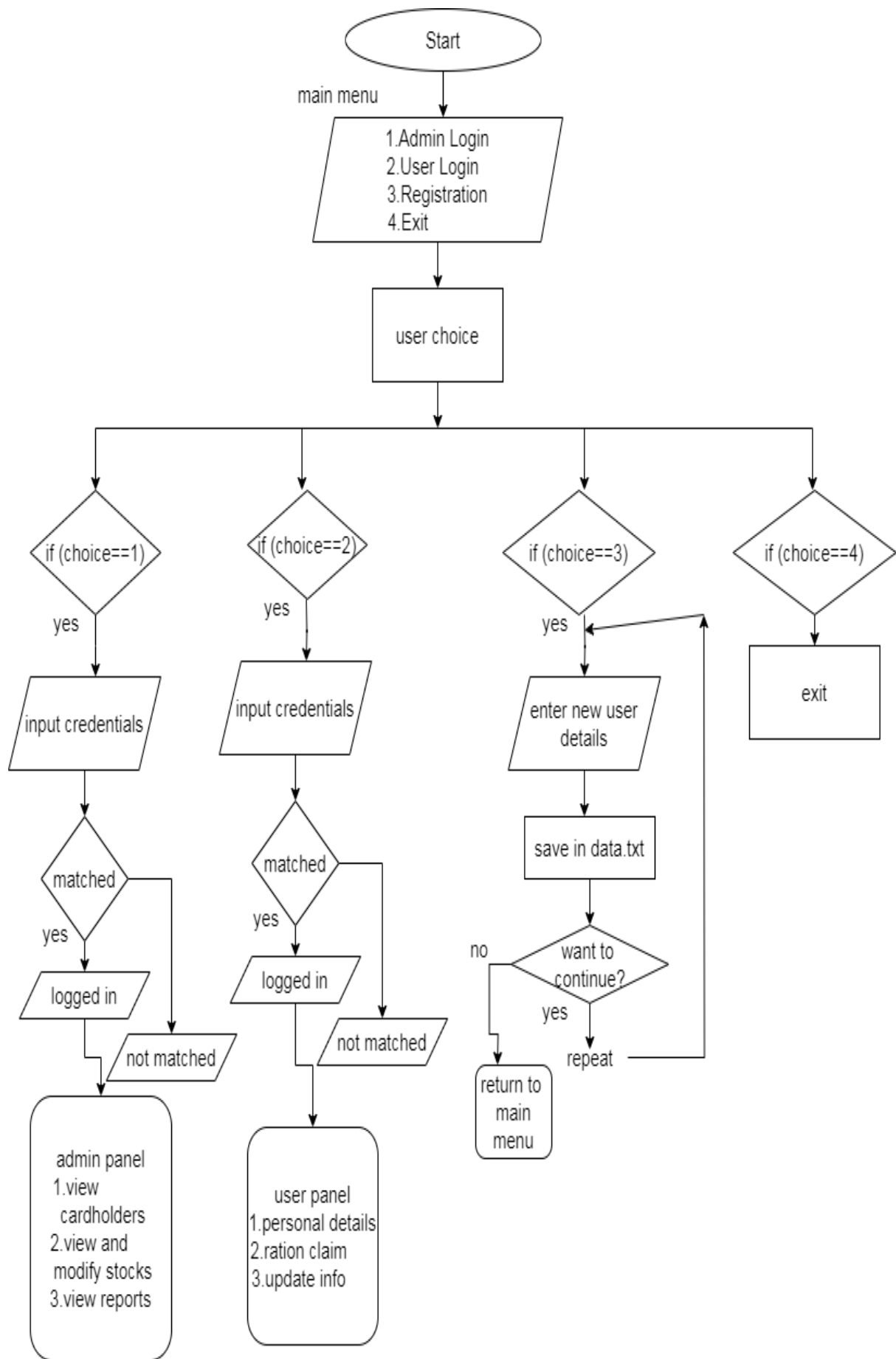
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1. Problem Description:-

Ration distribution systems are critical to delivering essential commodities like rice, wheat, and sugar to economically weaker sections of society. However, manual procedures at government outlets often result in long queues, record mismanagement, ration misuse, and data inconsistency. This project aims to design a C-based system that streamlines and secures the process of ration distribution. The system will allow both admin and user logins, register new ration cardholders, manage outlet stock, and record transactions securely using file storage. It minimizes the chances of fraud, automates stock tracking, and helps authorities generate detailed reports to monitor supply and usage. The goal is to bring transparency, efficiency, and accountability to the public distribution system through a console-based application.

1.1. Flow Diagram:





2. Modules for Project Implementation:

2.1 Module Description

<u>Module</u>	<u>Description</u>
<u>1. Admin Login</u>	-Authenticates government staff for access like <u>View System Reports</u> , <u>Initialize Stock</u> , <u>View Stock</u>
<u>2. User Login</u>	-Verifies existing cardholders using beneficiary and password and offers option like <u>View Card Information</u> , <u>View Monthly Ration</u>
<u>3. Registration Module</u>	-Registers new beneficiaries/user with personal data and initial ration setup like <u>Beneficiary Number</u> or <u>Card Number</u> , <u>Name</u> , <u>Mobile Number</u>
<u>4. Ration Claim Module</u>	-Enables users to claim their monthly quota and updates stock files.
<u>5. Reporting Module</u>	-Allows admins to view <u>stock reports</u> , <u>ration distributed</u> , and <u>remaining stock</u>

2.2 Platform Required:

Code::Blocks IDE With GCC Compiler:

Code::Blocks is a full-featured IDE (Integrated Development Environment) aiming to make the individual developer (and the development team) work in a nice programming environment offering everything he/she/they would ever need from a program of that kind.

GCC (GNU Compiler Collection) is a widely used, free, and open-source compiler suite that includes a compiler for the C programming language, often referred to as the "GNU C Compiler" or simply "gcc". GCC is a core component of the GNU Toolchain, a set of tools essential for developing applications and operating systems.

3. Screenshots of Project Output:

```
=====
      RATION DISTRIBUTION MANAGEMENT SYSTEM
=====
1. ADMIN LOGIN
2. USER LOGIN
3. REGISTER NEW CARDHOLDER
4. EXIT
-----
ENTER YOUR CHOICE <1,2,3,4>: 1

----- ADMIN LOGIN -----
ENTER ADMIN ID: admin
ENTER PASSWORD : 1234

>> WELCOME ADMIN!

===== ADMIN PANEL =====
1. VIEW SYSTEM REPORT
2. INITIALIZE STOCK
3. VIEW STOCK
4. LOGOUT
-----
ENTER YOUR CHOICE<1,2,3,4>: 1

===== SYSTEM REPORT =====
TOTAL REGISTERED CARDHOLDERS : 7
ESTIMATED RATIONS DISTRIBUTED:
    35 KG RICE
    14 KG WHEAT
    7 L OIL
CURRENT STOCK:
    200 KG RICE
    200 KG WHEAT
    200 L OIL
=====

===== ADMIN PANEL =====
1. VIEW SYSTEM REPORT
2. INITIALIZE STOCK
3. VIEW STOCK
4. LOGOUT
-----
ENTER YOUR CHOICE<1,2,3,4>: 2

--- SET STOCK QUANTITY ---
Enter rice <kg>: 1000
Enter wheat <kg>: 1000
Enter oil <lit>: 500
>> Stock saved successfully.
```

```

4. LOGOUT
-----
ENTER YOUR CHOICE<1,2,3,4>: 4

>> LOGGING OUT FROM ADMIN PANEL.

=====
      RATION DISTRIBUTION MANAGEMENT SYSTEM
=====
1. ADMIN LOGIN
2. USER LOGIN
3. REGISTER NEW CARDHOLDER
4. EXIT
-----
ENTER YOUR CHOICE <1,2,3,4>: 2

----- USER LOGIN -----
ENTER CARD NUMBER: 1000

>> WELCOME user1! LOGIN SUCCESSFUL.

===== USER PANEL =====
1. VIEW CARD INFORMATION
2. VIEW MONTHLY RATION
3. LOGOUT
-----
ENTER YOUR CHOICE<1,2,3>: 1

----- CARD INFORMATION -----
CARD NUMBER : 1000
NAME       : user1
MOBILE NO.  : 6824936593
-----

===== USER PANEL =====
1. VIEW CARD INFORMATION
2. VIEW MONTHLY RATION
3. LOGOUT
-----
ENTER YOUR CHOICE<1,2,3>: 2

----- MONTHLY RATION ALLOTMENT -----
RICE   : 5 KG
WHEAT  : 2 KG
OIL    : 1 LITRE
>> STOCK UPDATED.
-----

```



```
ENTER YOUR CHOICE<1,2,3>: 3

>> LOGGING OUT FROM USER PANEL.

=====
  RATION DISTRIBUTION MANAGEMENT SYSTEM
=====

1. ADMIN LOGIN
2. USER LOGIN
3. REGISTER NEW CARDHOLDER
4. EXIT
-----

ENTER YOUR CHOICE <1,2,3,4>: 3

----- REGISTER NEW CARDHOLDER -----
ENTER NEW CARD NUMBER: 9999
ENTER USERNAME      : xyz
ENTER MOBILE NUMBER <10 digits>: 3827998383

>> REGISTRATION SUCCESSFUL FOR CARD NUMBER 9999.

=====
  RATION DISTRIBUTION MANAGEMENT SYSTEM
=====

1. ADMIN LOGIN
2. USER LOGIN
3. REGISTER NEW CARDHOLDER
4. EXIT
-----

ENTER YOUR CHOICE <1,2,3,4>: 4

>> EXITING THE SYSTEM... THANK YOU!
```

4. Conclusion and Future Scope:

This project is a basic Ration Distribution Management System developed using the C programming language. It allows the admin to manage users and stock, and users to check their monthly ration. The system uses simple file handling to store and retrieve data. It shows how real-world problems like ration distribution can be managed using basic programming skills.

Future Scopes of this project are :-

- Add a module for Admin for deletion or updation of user records
- Limit the ration claiming once a month per user
- A particular sorting according to card number or name
- For security, binary files can be a good option
- Unique ID generation can be introduced for convenient registration
- A master file can be created for backup which will contain all the data.

5. References:

- I. **The C Programming Language – Brian W. Kernighan & Dennis M. Ritchie**
- II. **Computer Fundamental and Programming in C 2nd Edition – Reema Thareja**
- III. **TutorialsPoint – C Programming**
<https://www.tutorialspoint.com/cprogramming/index.htm>
- IV. **Programiz – Learn C Programming**
<https://www.programiz.com/c-programming>
- V. **YouTube – C programming tutorials**

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